Mastery in Data Analytics and Visualizations and Career Advancement

Lab sheet 04

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```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# 1. LOAD DATASET
df = pd.read_csv("superstore_sales.csv")

print("=" * 70)
print("ADVANCED DATA VISUALIZATION")
print("=" * 70)

# 2. DISPLAY DATA
print("\nFirst 5 Records:")
print(df.head())

print("\nDataset Dimensions:")
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])

# 3. BAR CHART - SALES BY REGION
region_sales = df.groupby("Region")["Sales"].sum().sort_values(ascending=False)
```

```
plt.figure(figsize=(8, 5))
sns.barplot(
    x=region_sales.index,
    y=region_sales.values
)

plt.title("Sales by Region")
plt.xlabel("Region")
plt.ylabel("Total Sales")
plt.show()

# 4. BAR CHART - SALES BY CATEGORY
category_sales = df.groupby("Category")["Sales"].sum().sort_values(ascending=False)

plt.figure(figsize=(8, 5))
sns.barplot(
    x=category_sales.index,
    y=category_sales.values
)

plt.title("Sales by Product Category")
plt.xlabel("Category")
plt.ylabel("Total Sales")
plt.xticks(rotation=20)
plt.show()

# 5. LINE CHART - MONTHLY SALES
```

```
df["Order_Date"] = pd.to_datetime(df["Order_Date"])

monthly_sales = (
    df.groupby(df["Order_Date"].dt.to_period("M"))["Sales"]
    .sum()
    .reset_index()
)
monthly_sales["Order_Date"] = monthly_sales["Order_Date"].dt.to_timestamp()

plt.figure(figsize=(12, 5))
sns.lineplot(
    data=monthly_sales,
    x="Order_Date",
    y="Sales",
    marker="o"
)

plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.xticks(rotation=45)
plt.show()

# 6. HISTOGRAM - SALES DISTRIBUTION
plt.figure(figsize=(8, 5))
sns.histplot(
    df["Sales"],
    bins=15,
```

```
    kde=True
)

plt.title("Distribution of Sales")
plt.xlabel("Sales")
plt.ylabel("Frequency")
plt.show()

# 7. BOX PLOT - SALES BY REGION
plt.figure(figsize=(8, 5))

sns.boxplot(
    data=df,
    x="Region",
    y="Sales"
)

plt.title("Sales Distribution by Region")
plt.xlabel("Region")
plt.ylabel("Sales")
plt.show()

# 8. SCATTER PLOT - SALES VS PROFIT
plt.figure(figsize=(8, 5))
sns.scatterplot(
    data=df,
    x="Sales",
    y="Profit",
```

```
    hue="Category"
)

plt.title("Sales vs Profit")
plt.xlabel("Sales")
plt.ylabel("Profit")
plt.show()

# 9. CORRELATION HEATMAP
numeric_columns = [
    "Sales",
    "Profit",
    "Quantity",
    "Discount"
]
correlation = df[numeric_columns].corr()

print("\nCorrelation Matrix:")
print(correlation)
plt.figure(figsize=(8, 6))
sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Heatmap")
```

```
plt.show()

# 10. SALES BY REGION AND CATEGORY
region_category = (
    df.groupby(["Region", "Category"])["Sales"]
    .sum()
    .reset_index()
)

plt.figure(figsize=(10, 6))

sns.barplot(
    data=region_category,
    x="Region",
    y="Sales",
    hue="Category"
)

plt.title("Sales by Region and Category")
plt.xlabel("Region")
plt.ylabel("Sales")
plt.show()

# 11. FINAL OBSERVATIONS
print("\n" + "=" * 70)
print("KEY OBSERVATIONS")
print("=" * 70)
```

```
print("\n1. Region with highest sales:")
print(region_sales.idxmax())
print("\n2. Category with highest sales:")
print(category_sales.idxmax())
print("\n3. Highest monthly sales:")
print(
   monthly_sales.loc[
      monthly_sales["Sales"].idxmax(),
      "Order_Date"
   ].strftime("%B %Y")
)
print("\n4. Correlation between Sales and Profit:")
print(correlation.loc["Sales", "Profit"])
print("\nVisualization completed successfully!")
```

Questions and Answers

1. Why is data visualization considered an essential component of data analytics?

Answer:

Data visualization converts raw data into graphical representations. It makes complex data easier to understand and helps identify trends, patterns, outliers, distributions, and relationships.

2. Differentiate between Matplotlib and Seaborn. Which library is more suitable for statistical visualizations and why?

Answer:

Matplotlib is a general-purpose visualization library that provides detailed control over graphs. Seaborn is built on Matplotlib and provides easier-to-use statistical visualizations with attractive default styles. Seaborn is more suitable for statistical visualizations because it provides functions specifically designed for statistical analysis.

3. Which type of chart would you choose to compare sales across different regions? Justify your answer.

Answer:

A bar chart is suitable for comparing sales across different regions because it clearly displays the sales value for each categorical region, making comparison easy.

4. What information can be obtained from a histogram and a box plot?

Answer:

A histogram shows the distribution and frequency of numerical data. A box plot shows the median, spread, quartiles, and possible outliers, and can be used to compare distributions between categories.

5. Explain the purpose of a scatter plot. How does it help in identifying relationships between variables?

Answer:

A scatter plot displays the relationship between two numerical variables. It helps identify patterns, trends, positive or negative relationships, and possible outliers.

6. What is a correlation heatmap? How can it assist in feature selection for machine learning?

Answer:

A correlation heatmap is a graphical representation of correlations between numerical variables. It helps identify strongly related features and can assist in selecting useful features while avoiding highly redundant variables.

7. Why is chart customization (titles, labels, legends, colors) important in data visualization?

Answer:

Chart customization improves readability and makes the information easier to understand. Titles, labels, legends, and appropriate colors help users correctly interpret the visualization.

8. What factors should be considered while selecting an appropriate visualization for a dataset?

Answer:

The type of data, number of variables, analytical objective, and relationship that needs to be shown should be considered. The selected chart should clearly communicate the required information and make patterns easy to identify.

9. How can misleading visualizations affect business decision-making? Give a real-world example.

Answer:

Misleading visualizations can make data appear different from reality and may cause incorrect business decisions. For example, using a distorted scale in a sales chart can make a small increase appear very large, causing management to overestimate sales growth.

10. Explain how data visualization supports storytelling and business intelligence in organizations.

Answer:

Data visualization converts complex business data into understandable graphical stories. It helps organizations identify trends, communicate findings, compare performance, and make informed business decisions.

OUTPUT

PS D:\PROGRAMS\Career advancement lab\lab sheet 4> python -u "d:\PROGRAMS\Career advancement lab\lab sheet 4\exp 4.py"

ADVANCED DATA VISUALIZATION

First 5 Records:

	Order_ID	Order_Date	Region	Category	Sales	Profit	Quantity	Discount
0	ORD-0068	2024-01-01	South	Technology	3490.29	-436.51	7	0.11

1 ORD-0033 2024-01-01 West Office Supplies 3485.59 1091.46 5 0.02

2 ORD-0027 2024-01-01 South Technology 3420.07 0.56 2 0.17

3 ORD-0020 2024-02-01 East Furniture 2725.30 129.27 9 0.19

4 ORD-0070 2024-02-01 Central Furniture 1198.92 112.32 10 0.39

Dataset Dimensions:

Rows: 80

Columns: 8

Correlation Matrix:

	Sales	Profit	Quantity	Discount
Sales	1.000000	0.014328	0.019414	-0.108738
Profit	0.014328	1.000000	-0.087141	0.000926
Quantity	0.019414	-0.087141	1.000000	-0.004322
Discount	-0.108738	0.000926	-0.004322	1.000000

KEY OBSERVATIONS

1. Region with highest sales:

West

2. Category with highest sales:

Furniture

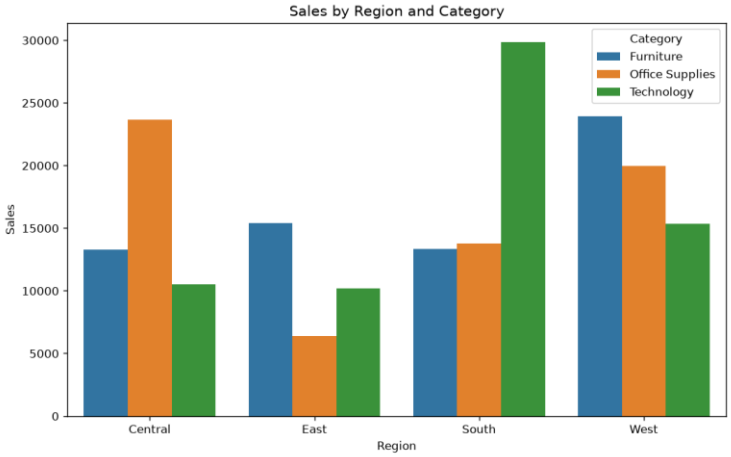
3. Highest monthly sales:

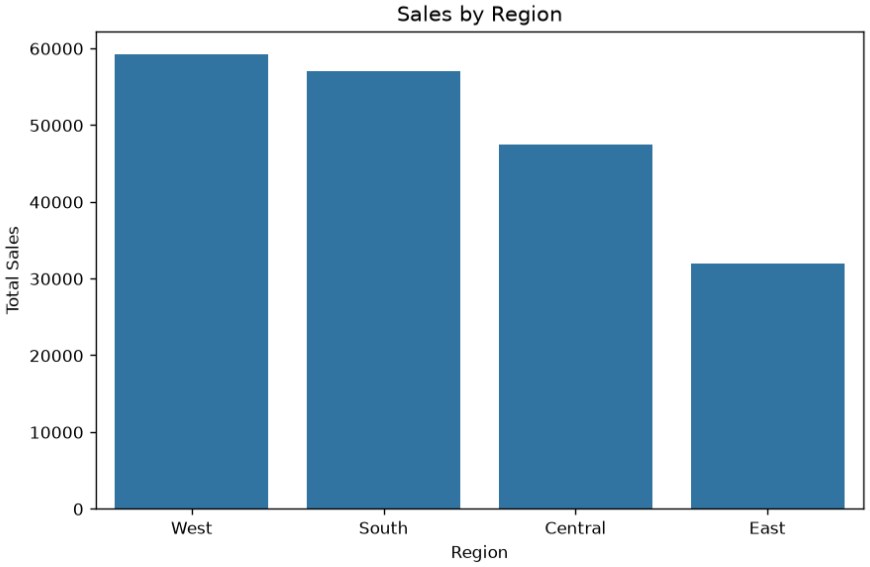
April 2024

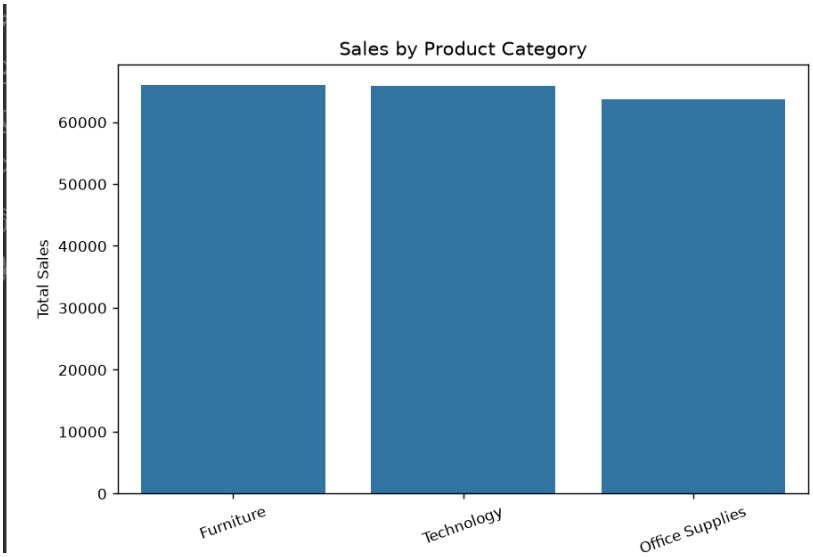
4. Correlation between Sales and Profit:

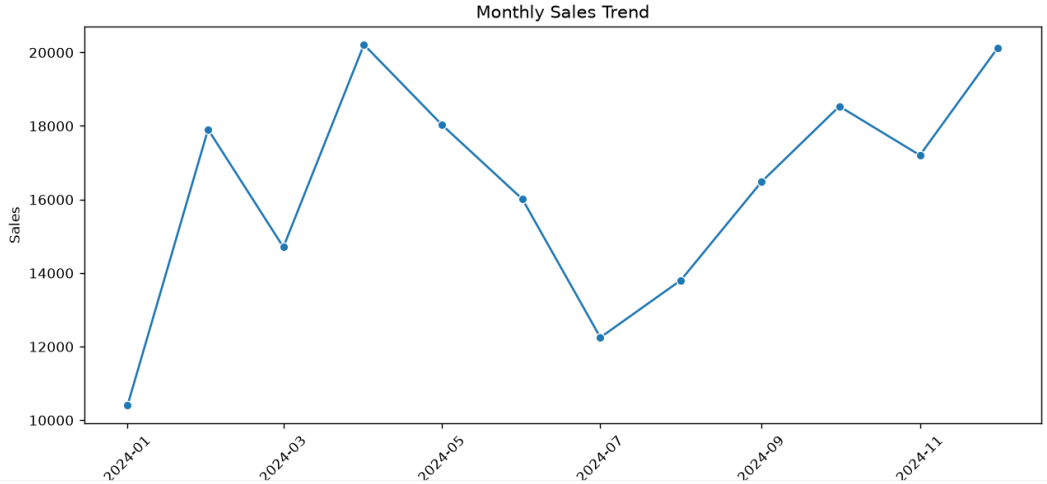
0.014328038030415057

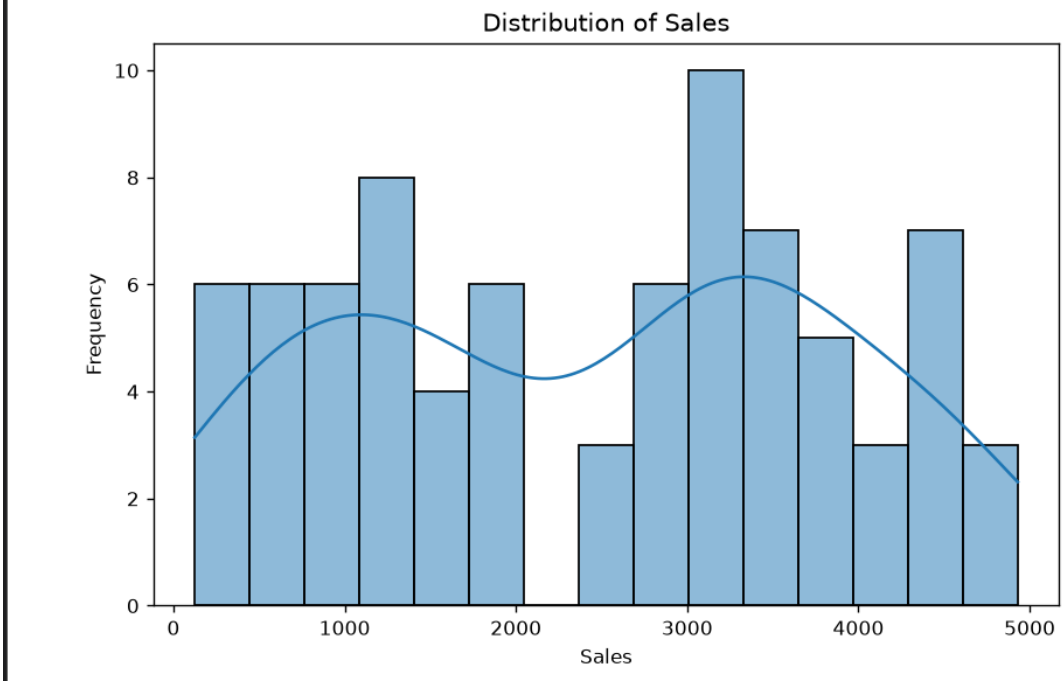
Visualization completed successfully!

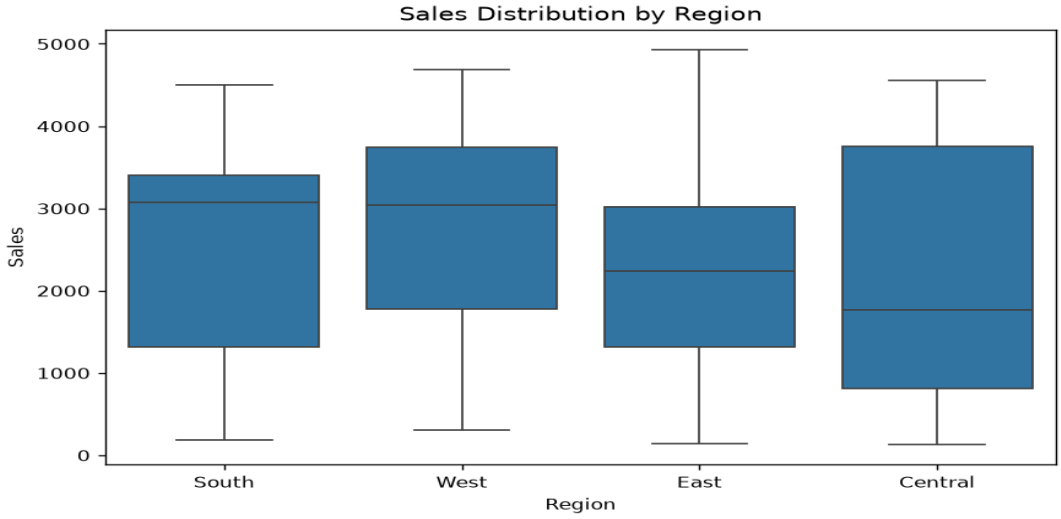

Sales by Region and Category
Category
Furniture
Office Supplies
Technology
Sales
30000
25000
20000
15000
10000
5000
0
Central
East
South
West
Region

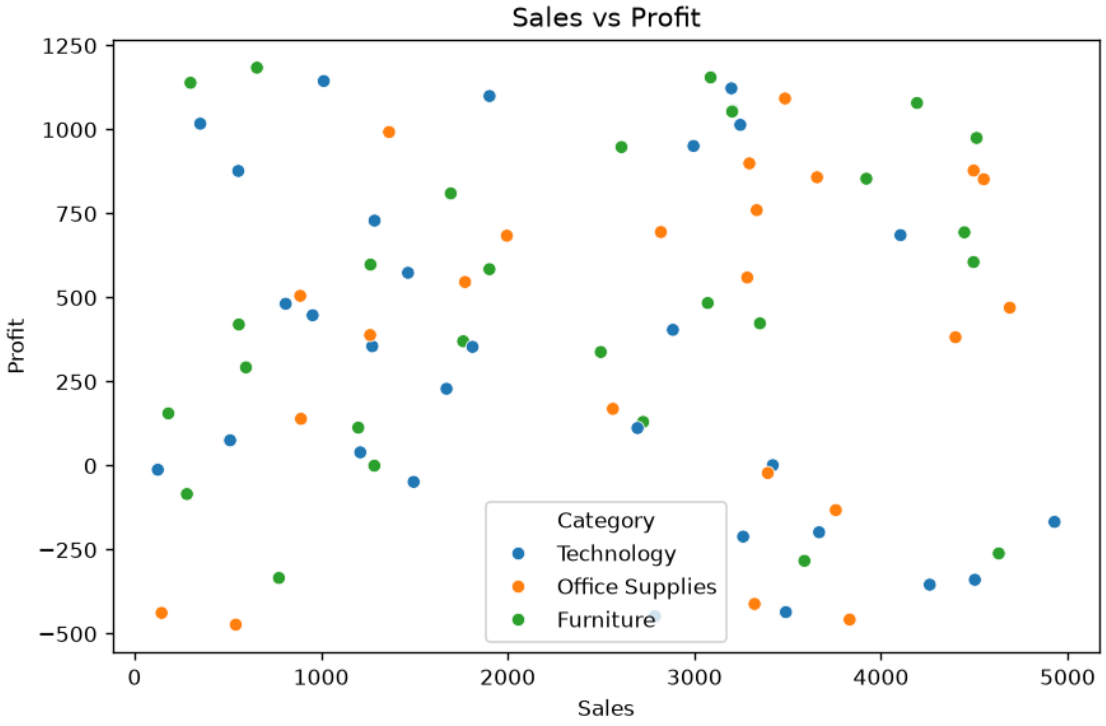

Sales by Region
Total Sales
60000
50000
40000
30000
20000
10000
0
West
South
Central
East
Region

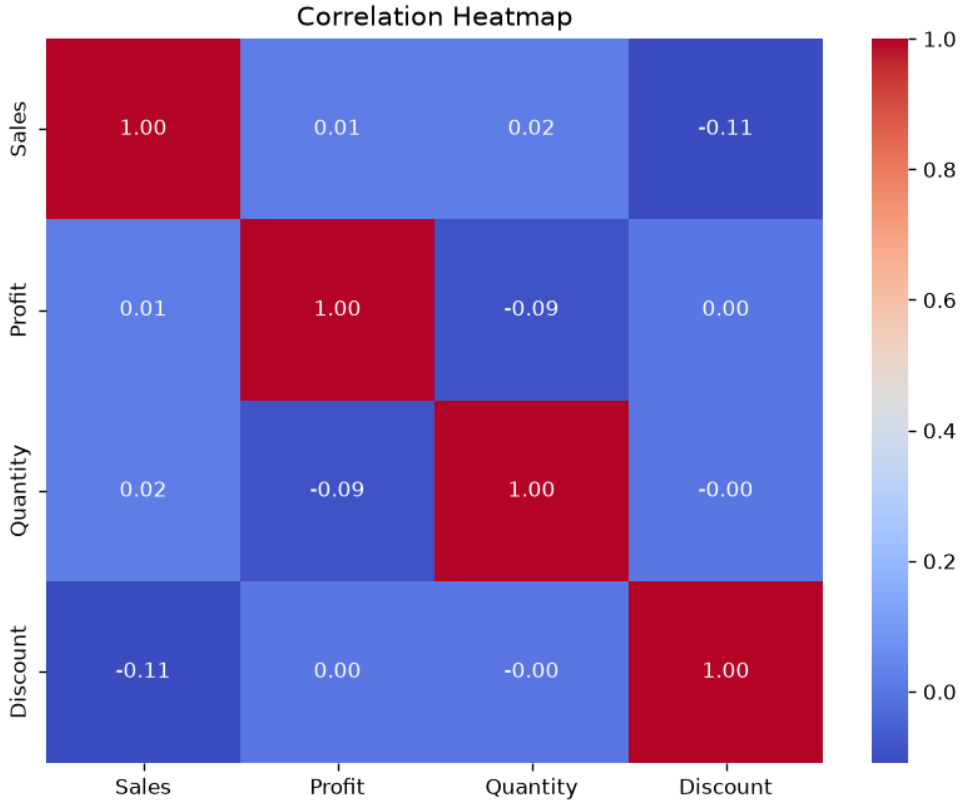

Sales by Product Category
Total Sales
60000
50000
40000
30000
20000
10000
0
Furniture
Technology
Office Supplies


Monthly Sales Trend
Sales
20000
18000
16000
14000
12000
10000
2024-01
2024-03
2024-05
2024-07
2024-09
2024-11


Distribution of Sales
Frequency
10
8
6
4
2
0
0
1000
2000
3000
4000
5000
Sales


Sales Distribution by Region
Sales
5000
4000
3000
2000
1000
0
South
West
East
Central
Region


Sales vs Profit
Profit
Sales
Category
Technology
Office Supplies
Furniture
1250
1000
750
500
250
0
−250
−500
0
1000
2000
3000
4000
5000


Correlation Heatmap
Sales
Profit
Quantity
Discount
1.00
0.01
0.02
-0.11
0.01
1.00
-0.09
0.00
0.02
-0.09
1.00
-0.00
-0.11
0.00
-0.00
1.00
1.0
0.8
0.6
0.4
0.2
0.0